

IMGAG longread meeting

17.12.2024

1st status update T2T

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Important Resources

- Koren et al. 2024: Gapless assembly of complete human and plant chromosomes using only nanopore sequencing (**Verkko**)
- Šikić et al. 2024: Telomere-to-telomere phased genome assembly using error-corrected Simplex nanopore reads (**Herro**)
- Li and Durbin 2024: Genome assembly in the telomere-to-telomere era (**Review**)
- ONT T2T Assembly Blogpost : <https://community.nanoporetech.com/t/ont-t2t-assembly-blogpost/10166>

Nanopore Read Types

- **Simplex**: Default ONT reads, LSK Library (R10.3)
- **Duplex**: Simplex reads, but with duplex chemistry (R10.3)
- **Herro**: Simplex or UL reads, corrected with ML algorithm (R10.3)
- **UL**: Special Library Prep, focus on max read length (R10.4)
- **PoreC**: Chromatin association
- **Polishing Kit**: Simplex reads, but with Polishing Kit chemistry

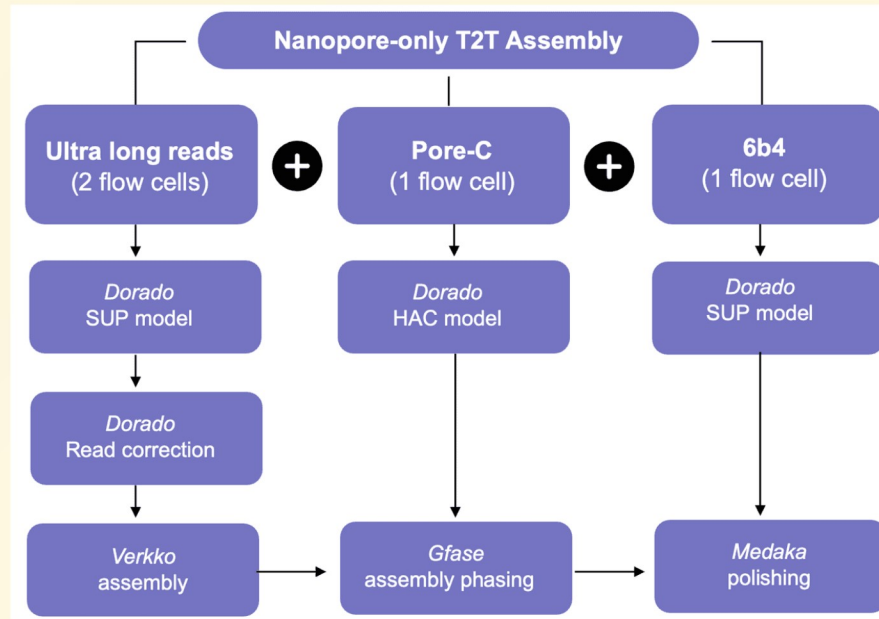
Public Datasets

Giab Sample **HG002** From Koren et al (2024):

- 15 FC Simplex R10
- Duplex reads called from simplex reads
- 8 FC R9.4 UltraLong
- PoreC

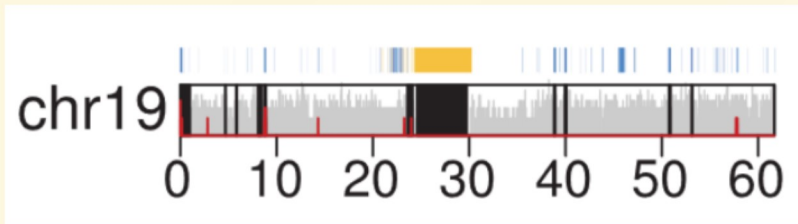
Published Pipelines

- ONT Recommended (Verkko with Herro)
- Verkko with Duplex (Koren et al. 2024)



Benchmarking with published data

Run on reduced dataset: Filtered on chr19

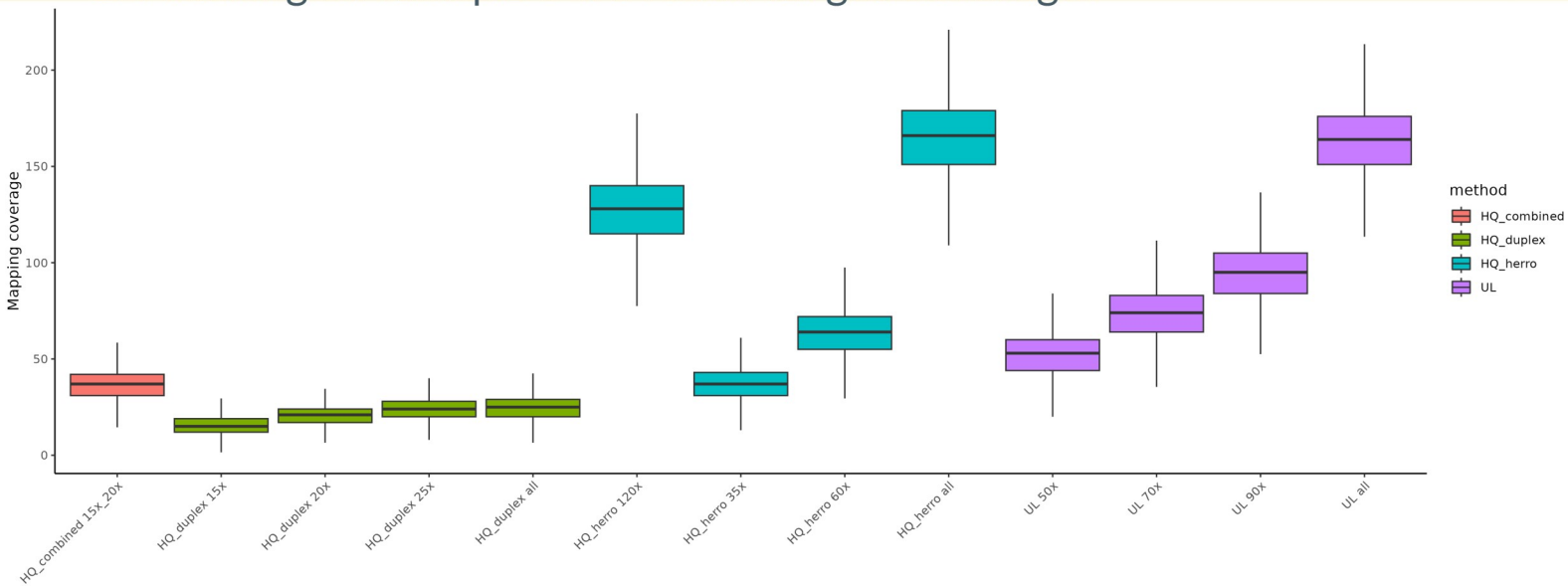


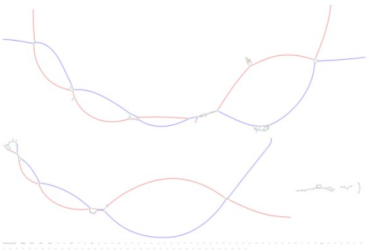
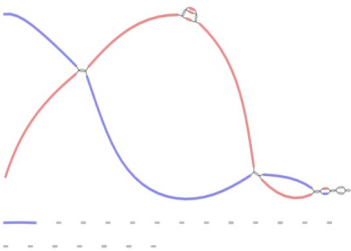
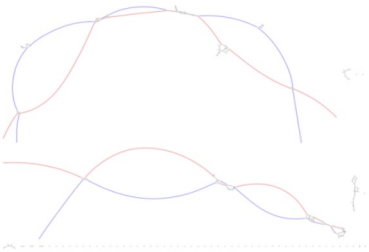
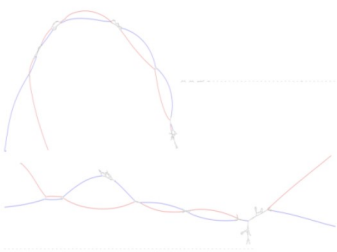
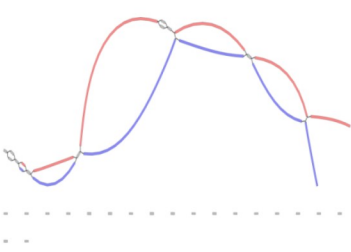
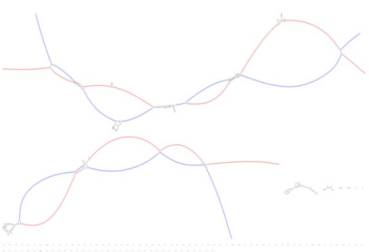
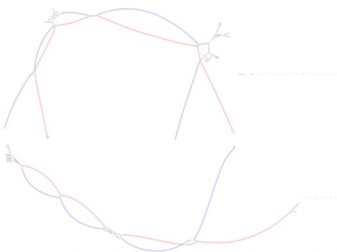
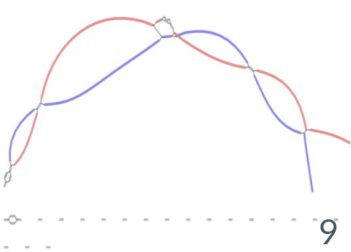
- *"Easy"* chromosome
- Fast computation

Benchmark Comparison Matrix

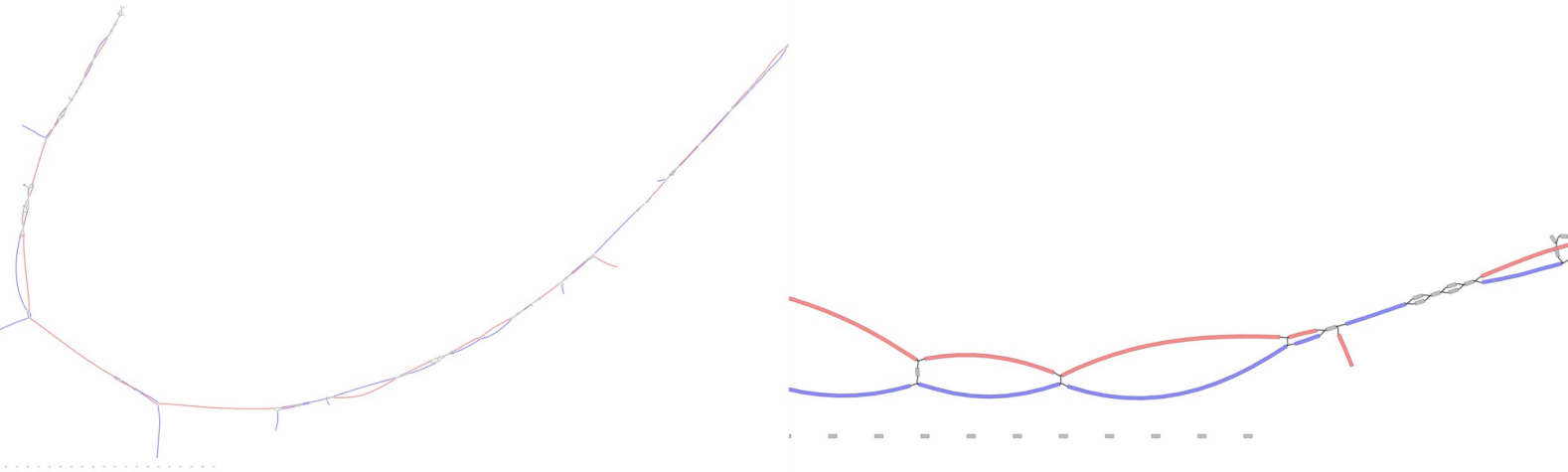
LR↓/UL→	50x	70x	90x
	-----	-----	-----
15x	duplex	--	--
20x	duplex	--	--
25x	duplex	--	--
35x	15x DU + 20x herro	--	--
35x	duplex/herro	duplex/herro	duplex/herro
60x	herro	herro	herro
120x	herro	herro	herro

1. Filtering on fastq. Do we reach target coverage?



	Herro 35x	Herro 60x	Herro 120x	Duplex 35x
UL 50x				
UL 70x				
UL 90x				

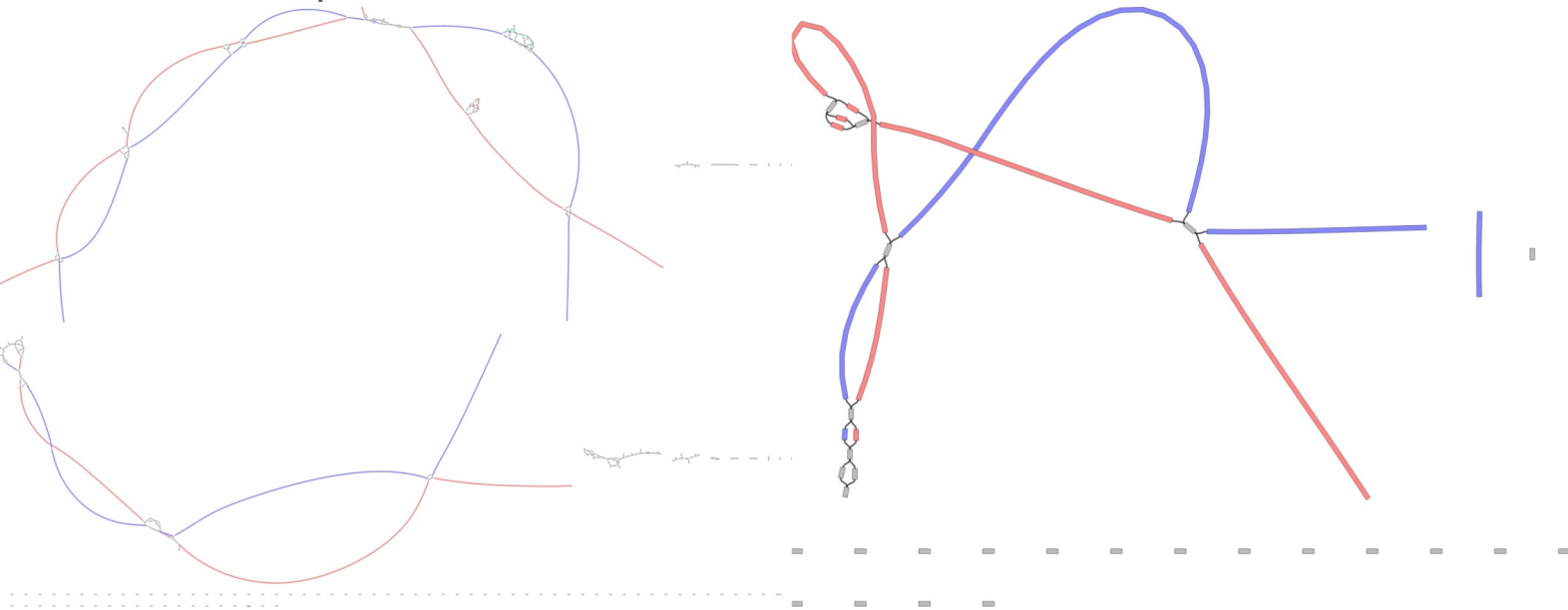
Left Duplex 15x, Right Duplex 20x, chr19



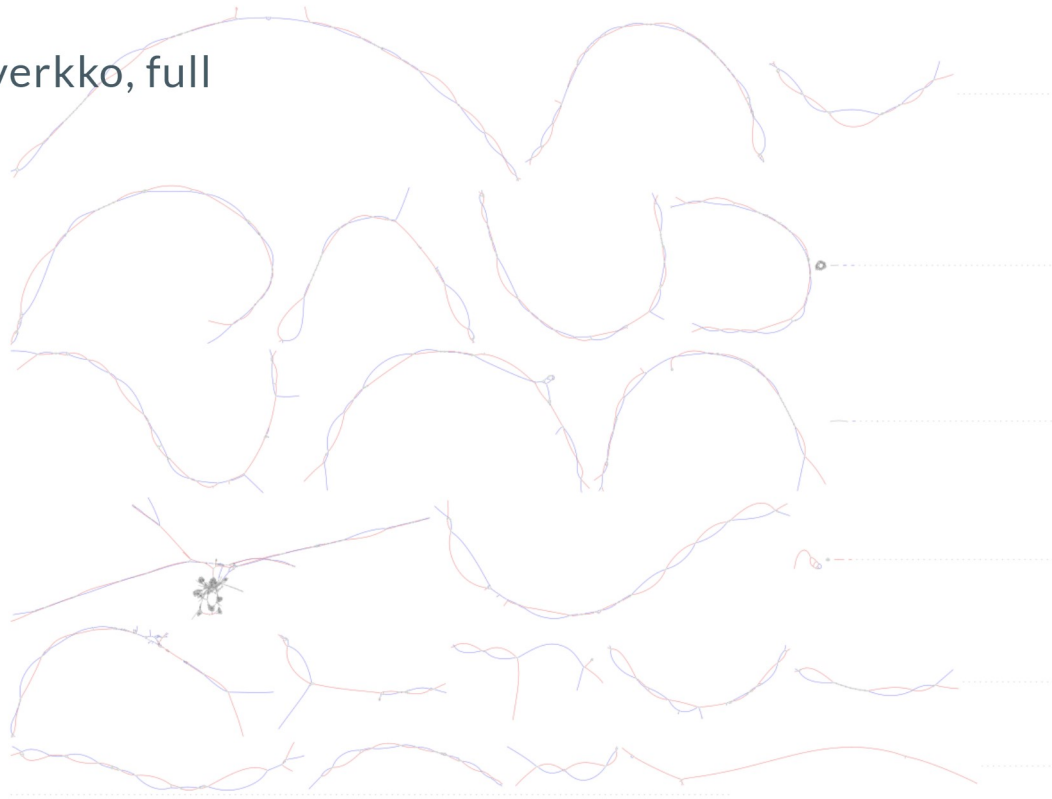
Key points:

- High coverages do not increase quality
- 35x HQ and 50x UL is optimal
- Low cov Duplex still ok

herro vs duplex



duplex, verkko, full



Completed T2T ?

no postprocessing, T2T detection with script, looking for TTAGGG

Duplex, UL70x, Verkko

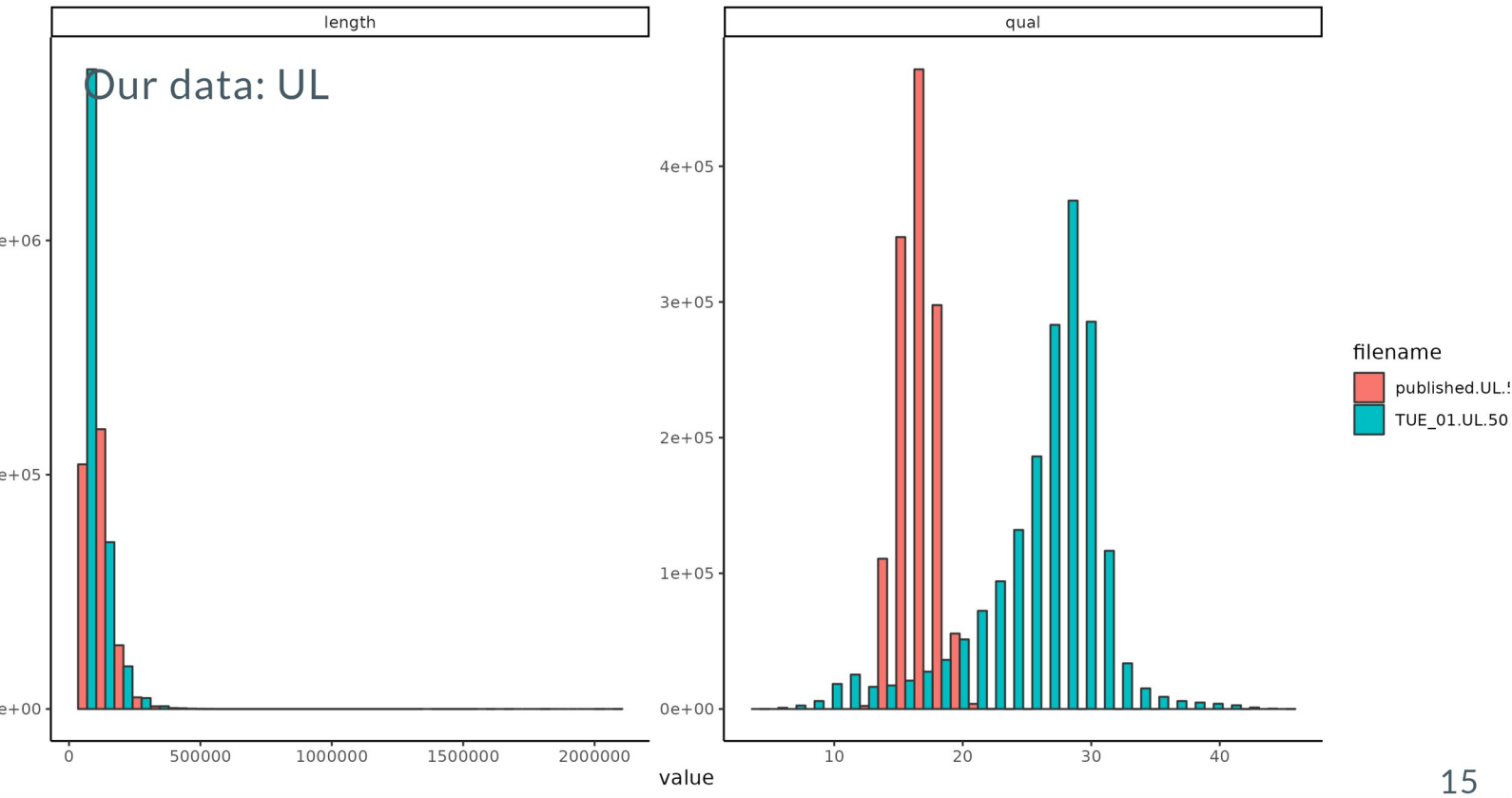
Haplotype1: **16** chromosomes T2T

Haplotype2: **14** chromosomes T2T

Duplex, UL70x, Gfase

Haplotype1: **8** chromosomes T2T

Haplotype2: **0** chromosomes T2T



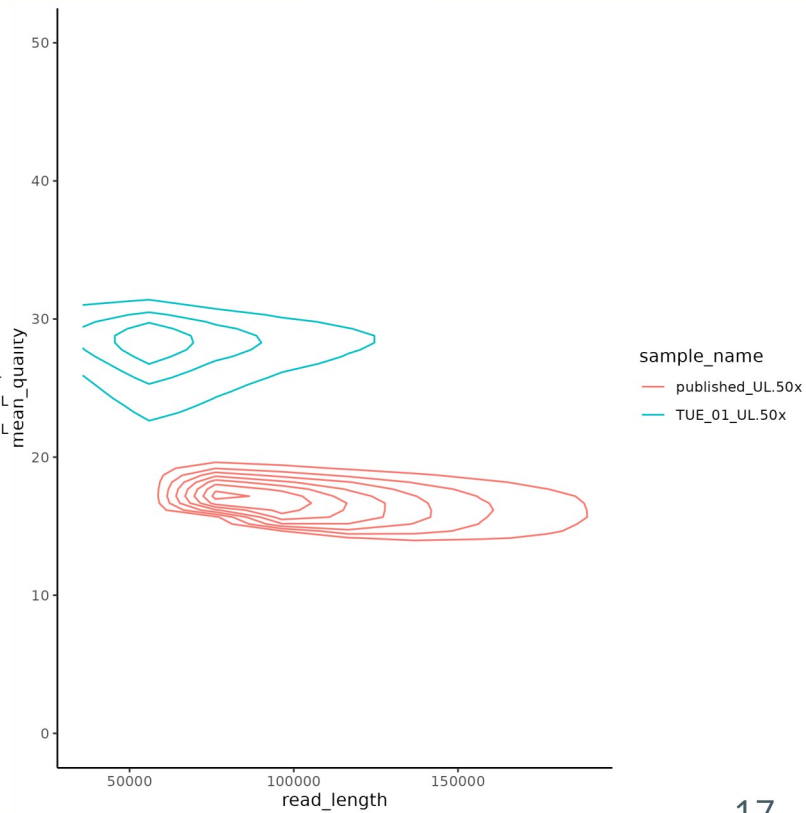
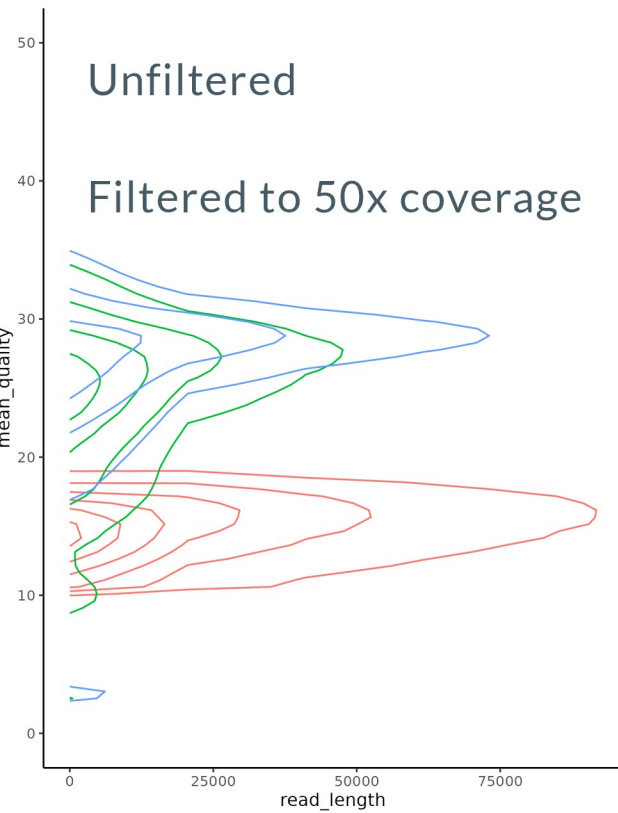
Our Data: Ultra-Long

Overview of the first two flowcells we sequenced.

Our data is unprocessed, published data "as downloaded"

filename	yield	N50	max readlen	yield>100kb	yield>200kb	reads > 1Mb
published.UL	496Gb	81748	1.9 Mb	190 Gb	31 Gb	45
run_04399.UL	114Gb	67652	1.7 Mb	38 Gb	10 Gb	29
run_04400.UL	108Gb	77925	2.0 Mb	40 Gb	9 Gb	54

Published data from 8 FC



After filtering:

filename	yield	N50	max readlen	yield>100kb	yield>200kb	reads > 1Mb
TUE_01.UL.50x	160GB	97739	2037kb	78GB	18GB	50
published.UL.50x	160GB	129072	1511kb	121GB	22GB	39

Target coverage can be reached with 2 good FC.

Additional reads for better read filtering are preferred.

4 additional FC with similar performance in analysis.

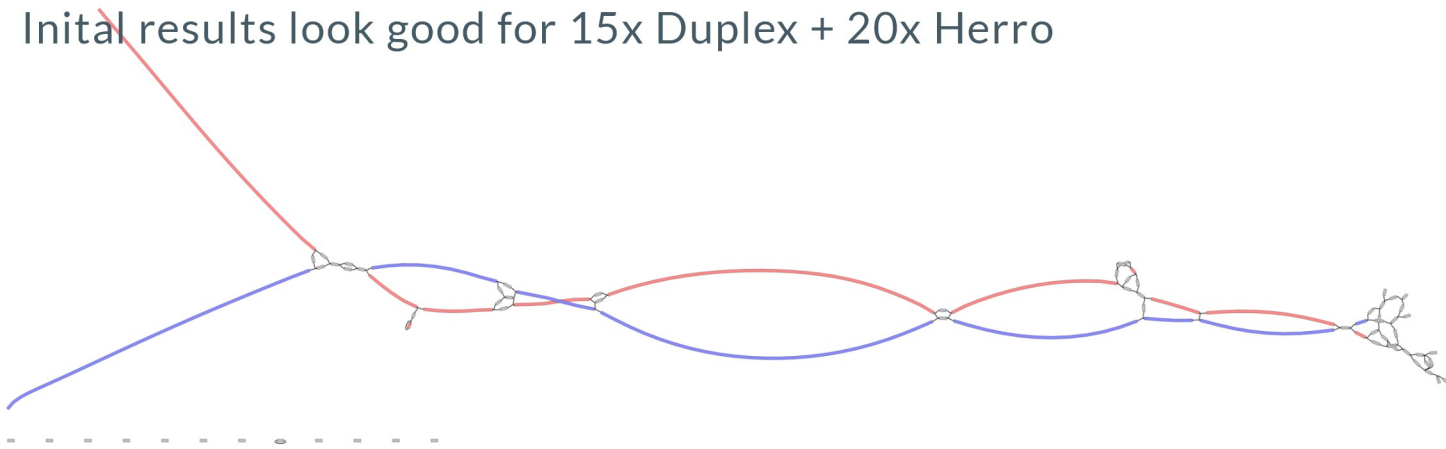
Our data: Duplex

Sample	Project	Type	Duplex Rate	Duplex Yield
23074LRa001	MYGENOME	LSK114	0.27	10Gb
23074LRa003	MYGENOME	LSK114	0.26	10Gb
24070LRa003	UL test	ULK114	0.01	0.6Gb
24070LRa010	UL test	ULK114	0.02	0.9Gb

Without Duplex optimizing. Published data reached ~15Gb / FC

Combine Herro and Duplex.

Initial results look good for 15x Duplex + 20x Herro



Discussion points

T2T requirement discussions:

- Genome completeness
 - How many T2T chromosomes do we need?
- Planned number of Flowcells
 - 2 or 3 UL ?
 - At least 45 GB Duplex (15x) --> 4 - 5 FC
 - 1 PoreC
 - 1 Polishing?

Discussion points

- Lab Establishment Priorities
 - UL : finished?
 - Duplex : required?
 - PoreC
 - Polishing Kit
- Bioinformatics Implementations
 - Downstream analysis of Genomes
 - Sequence HG002 for Benchmark?